Our final result (mid rapidity) and AABKL, arXiv:1411.2869 (forward rapidity)

Our result: `Remaining_RC_DGLAP_finite.pdf`, Eq. (8.1), with the evolution terms of `Checks_JIMWLK_BFKL.pdf`. AABKL: T. Altinoluk, N. Armesto, G. Beuf, A. Kovner, M. Lublinsky, "Single-inclusive particle production in proton-nucleus collisions at next-to-leading order in the hybrid formalism", their final result Eq. (4.22), the BK derivation Eqs. (5.31)–(5.39), and App. C. $a \equiv \frac{k^+}{\vee}$.

1 Answer

- **Term by term the two results do not coincide, and they should not.** The leading orders are different objects, so the NLO corrections are corrections to different things:
  - AABKL (forward): a collinear parton of the proton, with a PDF $x_p f(x_p)$, scatters eikonally; LO is $O(\alpha_s^0)$; NLO is one splitting of that parton.
  - Ours (mid rapidity): a gluon radiated by the colour charges ρ of a dilute projectile; LO is $O(\alpha_s)$ with $\langle\rho\rho\rangle$; NLO is one more gluon in $\Lambda < p^+ < \vee$.

  The finite constants are also scheme dependent: AABKL cut the soft gluon by an Ioffe time, we by sharp p^+ cutoffs.
- **Everything that must agree does agree** (Sec. 3): the target evolution, the rapidity bookkeeping, the final-state DGLAP term (same P_{gg}, same plus prescription, same kinematic limit, same scheme constant), the hadron convolution, and the $\beta_0/2$ part of P_{gg}.
- **The forward limit of our result has AABKL's structure** (Sec. 4): at fixed $a = \frac{k^+}{\vee} = O(1)$ the BFKL logarithm is $O(1)$, and only the target evolves, as in their Eq. (5.32).
- **What is new at mid rapidity** (Sec. 5): the BFKL evolution of the projectile over $\log \frac{\vee}{k^+} = \frac{1}{2}\log\frac{\vee}{\Lambda}$, which is as large as the target interval; the running of the LO coupling; and terms that exist only because the projectile is a set of colour charges.

2 The two final results side by side

Ours.

$$\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^2}\, D_{h/g}(\zeta,\mu_F)\, \frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO} + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO} + \mathcal{N}\Big[\log\frac{k^+}{\Lambda}\,\mathrm{JIMWLK}\otimes\mathrm{LO} + \log\frac{\vee}{k^+}\,\mathrm{BFKL}\otimes\mathrm{LO}\Big]\\
&\quad - \frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - \log\frac{\vee}{k^+}H_4 + \frac{\alpha_s N_c}{4\pi}\log(1-a)\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big] + \frac{d^3N_{V_\vee}}{d^3k} + \text{finite}.
\end{aligned} \tag{2.1}$$

AABKL, Eq. (4.22) (gluon channel; quark channels analogous), schematically:

$$\begin{aligned}
\frac{d\sigma}{d^2p_\perp d\eta} &= \int_{x_F}^1\frac{d\zeta}{\zeta^2}D(\zeta)\Big\{x_p f^g_{\mu^2}(x_p)\, s^A_{Y_T}[y,\bar{y}]\ (\mathrm{LO},\, O(\alpha_s^0))\\
&\quad + \frac{g^2}{(2\pi)^3}\int_0^{1-x_p}d\xi\,\frac{x_p}{1-\xi}f^g\Big(\frac{x_p}{1-\xi}\Big)\Big[\frac{1-\xi}{[\xi]_+}+\frac{\xi}{1-\xi}+\xi(1-\xi)\Big]2C_A\\
&\quad\times\Big\{\big[A^i(y-z)A^i(\bar{y}-z)-(A^iA^i)_{\mu^2}\big]\Big(s_A[y,\bar{y}]\ (\mathrm{PDF})+(1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar{y}]\ (\mathrm{FF})\Big)+\dots\Big\}
\end{aligned}$$

$$+\,\text{real and virtual terms with } s\,s\,s,\ \frac{1}{N_c^2}X,\ s_A \text{ (colour structures of the gluon channel)}$$

$$+\,\frac{g^2}{(2\pi)^3}x_p f^g(x_p)\int_0^1\frac{d\xi}{\xi}\int[A^i_\xi(y-z)-A^i_\xi(\bar y-z)]^2\Big[\frac{N_c^3}{N_c^2-1}(sss+sss)-\frac{1}{N_c^2}X-2C_A s_A\Big]\Big\}, \tag{2.2}$$

with the target evolved over Y_T, the projectile only over $Y_P=\ln\frac{1}{x_p}+O(1)$ (their Eq. (2.11)), and the PDFs and FFs in their scheme, $f_{\mu^2}=f^{\overline{MS}}_{1.26\mu^2}$. The last line is the soft-gluon term with the Ioffe-time cutoff in A_ξ; they stress it is not an extra evolution, since its "evolution interval" depends on the momentum of the emitted gluon.

3 What agrees

AABKL	ours	how it was checked
target evolution: BK for the dipole, from $s_0\frac{\partial}{\partial s_0}\sigma=0$, Eqs. (5.32)–(5.39)	$\log\frac{k^+}{\Lambda}$ JIMWLK $\otimes$ LO, from the independence of Λ	our two-ρ terms $\propto\log\frac{k^+}{\Lambda}$ are JIMWLK$\otimes$LO exactly (154 colour structures); BK is its large-N_c dipole form. Same derivation: cutoff independence
projectile interval $Y_P=\ln\frac{1}{x_p}$, Eq. (2.11)	$\log\frac{\vee}{k^+}$	$x_p\leftrightarrow\frac{k^+}{\vee}$ (largest projectile momentum $\leftrightarrow P^+$)
FF (after-target) term: P_{gg} with the plus prescription at the soft end, $\int_0^{1-x_p}d\xi$, $\log\big((y-\bar y)^2\mu^2\big)$, App. C	DGLAP term $-\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}\mathrm{LO}_\xi[\ell_\Delta(\mu_F)]$ from Group II Row IV	with $\xi_{\mathrm{AABKL}}=1-\xi$: $P_{\mathrm{ours}}(\xi)-2N_cP_{\mathrm{AABKL}}(1-\xi)=0$; same limit $(1-x_p\leftrightarrow 1-a)$; $\ell_\Delta=\log\frac{(y'-y)^2\mu^2}{c_0^2}$ (`dglap_rc/checks.json`)
scheme: $f_{\mu^2}=f^{\overline{MS}}_{1.26\mu^2}$, $R=2e^{\Psi(1)}$, Eq. (7.22)	$\overline{MS}$ with $c_0=2e^{-\gamma_E}$	$R=c_0=1.12292$, $R^2=c_0^2=1.26095$ exactly (`checks.json`)
$\frac{\beta_0}{2}\delta(1-\xi)$ of P_{gg}: from their virtual term, $\int_0^1 d\xi\big[\frac{1-\xi}{[\xi]_+}+\frac{\xi}{[1-\xi]_+}+\xi(1-\xi)\big]=-\frac{11}{6}$, $2C_A\cdot\frac{11}{6}=\beta_0$	from the coupling of LO (A3 and the shockwave through the loop cancel, `Steps_line1_and_mu.pdf`)	both complete $P^{(r)}_{gg,+}$ to the full P_{gg}, which is absorbed in $D_{h/g}(\mu_F)$
hadron convolution $\int_{x_F}^1\frac{d\zeta}{\zeta^2}D(\zeta)$	$\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^2}D_{h/g}(\zeta)$	same form
colour structures of the one-gluon real/virtual terms (sss, X/N_c^2, s_A)	$\mathrm{Tr}\{T^eAT^eB\}$ traces	same colour algebra; the kernels differ (emission by charges)

4 The forward limit of our result

AABKL (their Eq. (5.31)): "production at fixed x_p but at higher energy". For us that is $\vee\to\vee e^\Delta$ at fixed $a=\frac{k^+}{\vee}$, i.e. $k^+\to k^+e^\Delta$. Then $\log\frac{\vee}{k^+}=\log\frac{1}{a}$ is fixed and $\log\frac{k^+}{\Lambda}$ grows by Δ. From Eq. (2.1),

$$\frac{\partial}{\partial\log\vee}\frac{d^3N_g}{d^3k}\bigg|_a=\mathcal{N}\,\mathrm{JIMWLK}\otimes\mathrm{LO}+O(\alpha_s^2), \tag{4.1}$$

only the target evolves. This is their Eq. (5.32): the energy dependence at fixed x_p is the evolution of the target (s_{Y_T}), BK in their large-N_c dipole form, Eq. (5.39). In this limit our $\log\frac{\vee}{k^+}\,\mathrm{BFKL}\otimes\mathrm{LO}=\log\frac{1}{a}\,\mathrm{BFKL}\otimes\mathrm{LO}$ is an $O(\alpha_s)$ term without a large logarithm, like their projectile evolution over $Y_P=\ln\frac{1}{x_p}=O(1)$, which they absorb into the PDF at the scale of the projectile (their Sec. III.A). So at leading logarithm the forward limit of our result is AABKL's structure.

5 What is different, and why

- **BFKL of the projectile.** At mid rapidity $k^+=\sqrt{\vee\Lambda}$, so $\log\frac{\vee}{k^+}=\log\frac{k^+}{\Lambda}=\frac{1}{2}\log\frac{\vee}{\Lambda}$: the projectile interval is as large as the target one and must be resummed. It is $\log\frac{\vee}{k^+}\,\mathrm{BFKL}\otimes\mathrm{LO}$, your App. E formula. AABKL say it themselves: "For small values of x one indeed has to take into account the P^+ dependence" of the projectile distribution (Sec. III.A, item 3). Forward it is absent because $\ln\frac{1}{x_p}=O(1)$.

- **Running coupling.** Our LO is $\propto g^2$, so its coupling is renormalized: $\frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\mathrm{LO}$ and the constant $\frac{\alpha_s}{2\pi}N_c(\frac{67}{9}-\frac{\pi^2}{3})\mathrm{LO}$. AABKL's LO is $O(\alpha_s^0)$; they have no such term, and β_0 appears only in P_{gg}.

- **No PDF DGLAP.** Our projectile is ρ, not a collinear parton: the parent gluon is emitted at distance $|x - w|$ from a charge, so the before-target emission (Group II Row I) is finite. AABKL have a PDF term $\propto P \otimes f \log \mu^2$.
- **Terms that exist only for a projectile of colour charges:** the linear-polarization term ($\frac{k^i k^j}{k^2}$, line 3 of Group II Row IV), emission of the LO gluon after the target by the rotated charge, and the three- and four-ρ terms.
- **The limit $\xi \geq \frac{k^+}{\vee}$ in our virtual term** (Group I Row II) gives the pole Q1 $\propto \log(1-a)$ and $V_\vee = -\frac{\alpha_s}{2\pi} N_c \log(1-a) \log(\frac{1}{a} - 1)$LO. AABKL's virtual term has $\int_0^1 d\xi$ and neither term. Both vanish at mid rapidity in the high-energy limit ($a \to 0$). With AABKL's choice $\int_0^1$ they would be absent from ours too.
- **Quark channels** (C_F terms, their Eq. (4.21)): only in AABKL; we have $n_f = 0$.
- **Finite constants are scheme dependent:** the Ioffe-time cutoff vs. sharp cutoffs moves $O(\alpha_s)$ terms between "evolution" and "finite" (their Eqs. (3.11)–(3.13) and their last term in Eq. (4.22)). A numerical agreement of the finite parts is therefore not expected even where the structures are the same.

6 Verdict

Our result is consistent with AABKL where it has to be: the same target evolution and rapidity bookkeeping (forward limit = their structure), the same final-state DGLAP term in the same scheme, and the same P_{gg} including its $\beta_0/2$ part. It differs where it must: at mid rapidity the projectile side has its own large logarithm (BFKL), our LO carries a coupling that runs, and a projectile of colour charges produces terms a collinear parton does not. A term-by-term match is not possible because the two leading orders, and the two soft-gluon regularizations, are different.